Stochastic Attention via Langevin Dynamics on the Modern Hopfield Energy

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Motivation

Attention heads are deterministic: the same query always returns the same softmax-weighted memory average. Yet many tasks require *generating* novel, plausible patterns from structured memory.

We show that the gradient of the **modern Hopfield energy** is exactly the identity minus the attention map, and that applying **Langevin dynamics** to this energy yields *stochastic attention*: a **training-free sampler** controlled by a single inverse temperature β.

$\beta \to \infty$: exact retrieval	$\beta \approx \beta^*$: **generation of novel patterns**	$\beta \to 0$: isotropic noise

No score network, no training loop, no contrastive objective. Because the score $\nabla \log p_\beta = \beta(\mathbf{T} - \boldsymbol{\xi})$ is the attention map itself, any pretrained attention head becomes a zero-shot stochastic sampler.

Method: Stochastic Attention Update

Setup. $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{N \times K}$ stores K patterns. Sample from the Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ using ULA:

$$\boldsymbol{\xi}_{t+1} = \underbrace{(1-\alpha)\boldsymbol{\xi}_t}_{\text{contraction}} + \underbrace{\alpha \mathbf{X} \operatorname{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}_t)}_{\text{attention pull to memories}} + \underbrace{\sqrt{\frac{2\alpha}{\beta}}\boldsymbol{\epsilon}_t}_{\text{Gaussian noise}}$$

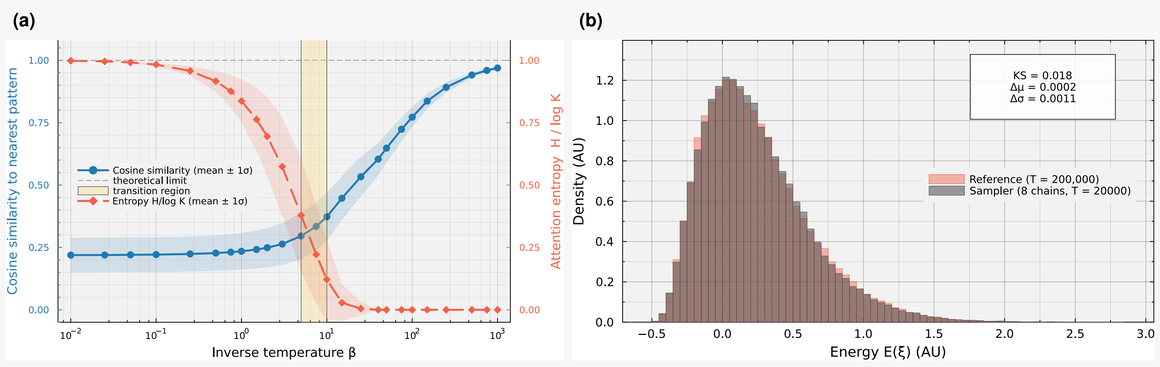
Step cost: $\mathcal{O}(NK)$ — same as one attention head.
No learned parameters. $\mathbf{X}$ is fixed; no training required.
Convergence (convex regime, $\beta\sigma_{\max}^2 < 2$): W_2-geometric convergence; MALA acceptance = 99.2% at $\alpha = 0.01$ confirms negligible ULA bias.
All $\beta > 0$: the continuous-time SDE is geometrically ergodic (dissipativity of E).

Temperature selection via entropy inflection. The attention entropy $H(\beta)$ transitions from $\log K$ (uniform) to 0 (hard attention). Setting $d^2H/d\beta^2 = 0$ identifies the retrieval–generation boundary β^*. For random unit-norm memories:

$$\beta^* \sim \sqrt{N}, \qquad \mathrm{SNR}^* = \sqrt{\frac{\alpha\beta^*}{2N}} \sim N^{-1/4}.$$

For structured data (e.g., protein sequences with conserved residues), evaluate the inflection numerically.

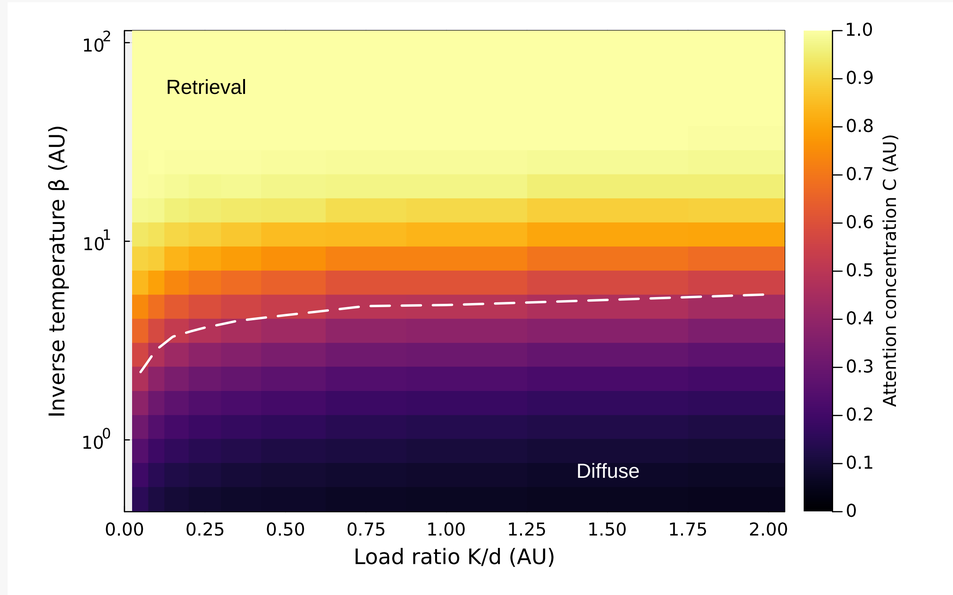
Experiment 1 — Phase Behavior & Convergence (Synthetic)



(a) $d = 64$, $K = 16$. Cosine similarity (blue) and scaled entropy (coral) exhibit a smooth sigmoidal transition centered at $\beta \approx 5$–10 (gold band), consistent with the predicted $\beta^* \sim \sqrt{N}$.

(b) $d = 8$, $K = 4$, $\beta = 5$. Eight independent chains (gray) converge to a long-run reference (coral); KS statistic confirms the correct Boltzmann target.

Experiment 2 — Load Ratio Phase Diagram (Synthetic)



Attention concentration $C = 1 - H(\mathbf{a})/\log K$ over load ratio K/d and β ($d = 64$). The $C = 0.5$ phase boundary (dashed) shifts right with increasing β: higher temperature extends reliable retrieval to larger memory loads, consistent with classical Hopfield capacity theory.

Experiment 3 — MNIST Image Generation ($K = 100$, $d = 784$)

Setup. 100 images of digit "3"; 30 independent chains; 150 generated samples. $\beta = 2000$ (SNR=0.113, retrieval) and $\beta = 200$ (SNR=0.036, generation).

Method	Novelty↑	Diversity↑	Energy↓
Bootstrap	0.000	0.459	−0.500
Gaussian perturbation	0.004	0.450	−0.496
Random convex combo	0.092	0.008	−0.399
GMM-PCA	0.198	0.419	−0.303
VAE (latent = 8)	0.214	0.441	−0.286
DDPM†	0.938	0.991	+0.44
MALA ($\beta = 2000$)	0.151	0.598	−0.305
SA ($\beta = 2000$, retrieval)	0.152	0.600	−0.303
SA ($\beta = 200$, generation)	**0.548**	**0.885**	—

†DDPM produces isotropic noise (max-cos = 0.062) despite 5,000 training epochs.

SA at $\beta = 200$ achieves **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while matching the Metropolis-corrected MALA gold standard.

Experiment 4 — Protein Sequence Generation (Pfam RRM, $K = 68$, $d = 59$)

Setup. 68 aligned RRM sequences; one-hot encoded, PCA-reduced (95% variance). Entropy inflection gives $\beta^* = 3.85$ (half the random-pattern prediction $\sqrt{d} = 7.68$; conserved residues increase effective similarity variance).

Method	Novelty↑	Diversity↑	SeqID	KL↓
Bootstrap	0.000	1.000	0.645	0.133
VAE (latent = 8)	0.621	0.834	0.532	0.416
SA ($\beta = 77$, retrieval)	0.243	0.991	0.616	0.107
SA ($\beta = 8$, generation)	**0.623**	**1.001**	0.538	**0.060**

SA matches the VAE on novelty while achieving **6.9**× lower amino acid composition divergence (KL = 0.060 vs. 0.416). The training-free Langevin score preserves family-level fidelity that learned models cannot maintain at small K.

Limitations & Future Work

ULA bias scales with step size α; MALA (with Metropolis correction) is preferred at larger α.
Inter-basin mixing at high β is exponentially slow; the multi-chain protocol provides a practical workaround, but quantitative scaling laws remain open.
Fixed memory: $\mathbf{X}$ must be known at sampling time; streaming or latent memory settings require extension.
Generation fidelity at high temperature yields structured interpolations rather than crisp samples; operating near β^* improves quality.
Future: multi-head attention, hierarchical memories, retrieval-augmented generation, in-context learning.

This work grew out of CHEME-5820: Machine Learning & AI Methods for Engineers, Cornell University.

github.com/varnerlab/stochastic-attention-study-paper